Yuulin An

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Department of Economics, Colorado State University
1771 Campus Delivery, Fort Collins, CO 80523

RESEARCH INTERESTS & SKILLS

Fields Health, Environmental, Demographic Economics
Software R, Python, Stata, Git/GitHub, LaTeX
Causal Inference Difference-in-Differences, Event Studies, Regression Discontinuity, Instrumental Variables
Language Khmer (native), English (fluent)

EDUCATION

Colorado State University — Fort Collins, CO
Ph.D., Economics (Candidate) — Expected Spring 2027
M.A., Economics — May 2024

Coe College — Cedar Rapids, IA
B.A., Economics and Business Administration (Cum Laude) — May 2022

JOB MARKET PAPER

Airborne Lead and Birth Outcomes: Evidence from Leaded Fuel in Auto Racing [PDF]

Abstract: The National Association for Stock Car Auto Racing (NASCAR) and Automobile Racing Club of America (ARCA) races continued to use leaded gasoline until 2007, despite the removal of leaded gasoline from on-road vehicles long ago. Exploiting this natural experiment, I study how airborne lead exposure affected birth outcomes using administrative birth records linked to race events at Texas Motor Speedway (TMS) in Fort Worth, Texas. Before the transition to unleaded fuel, mothers living within 25 km of TMS experienced lower infant birthweight, shorter gestation, and a higher probability of low birthweight associated with race exposure, while the corresponding birthweight and gestational-age estimates are near zero after the transition. To account for how lead accumulates in the body, I developed a pregnancy-specific physiologically based pharmacokinetic (PBPK) model that simulates maternal and fetal daily blood lead trajectories. I find that moving from the 25th to the 75th percentile of simulated blood lead in the second trimester is associated with an approximately 73-gram reduction in birthweight. Suggestive evidence points to gestational hypertension as a mechanism through which lead shortens gestation. These findings suggest that even a relatively low level of lead can have a meaningful effect on maternal and infant health.

PEER-REVIEWED ARTICLES

An, Y., Chin, S. & Miller, R. (2025). Growing Old in Rural America: Measuring Late Life Health and Economic Well-being. *The Journal of the Economics of Ageing*, 31, 100565.

WORKING PAPERS

Population Aging, Local Economic Growth, and the Urban-Rural Divide, with Tim Komarek and Ray Miller. Under Review at *Regional Science and Urban Economics*.

Maternal Lead Exposure and Birth Outcomes: Evidence from Natural Experiments in the Phase-out of Leaded Gasoline, with Sammy Zahran, Christopher Keyes, Sayorn Chin, and Susan Averett.

SELECTED WORKS IN PROGRESS

When Football Comes to Town: Provider Decision-Making and Delivery Complications, with Sayorn Chin and Sammy Zahran.

Urban Heat Island and Health, with Tim Komarek, Sammy Zahran, and Sayorn Chin.

OTHER PUBLICATIONS

Population Aging and Economic Growth in Colorado: Why Regional Connectivity Matters for Rural Communities, with Ray Miller and Tim Komarek. *Regional Economic Development Institute*, March 2026.

TEACHING EXPERIENCE

Department of Economics, Colorado State University Fort Collins, CO
Graduate Teaching Instructor Aug. 2025 - present

- Spring 2027: Introduction to Econometrics (1 section)
- Spring 2027: Working with Data (1 section)
- Summer 2026: Working with Data (1 section)
- Spring 2026: Working with Data (2 sections)
- Fall 2025: Working with Data (2 sections)
- Summer 2025: Working with Data (1 section)

Department of Economics, Colorado State University Fort Collins, CO
Graduate Teaching Assistant Aug. 2022 - 2025

- Spring 2025: Principles of Macroeconomics (3 sections)
- Fall 2024: Principles of Microeconomics (3 sections)
- Spring 2024: Principles of Microeconomics (3 sections)
- Fall 2023: Principles of Macroeconomics (3 sections)
- Spring 2023: Principles of Microeconomics (3 sections)
- Fall 2022: Principles of Microeconomics (3 sections)

RESEARCH EXPERIENCE

Regional Economic Development Institute Fort Collins, CO
Research Assistant to Dr. Ray Miller Summer 2025

Future of Work, Russel Sage Foundation Fort Collins, CO
Research Assistant to Dr. Zachary Schaller Summer 2024

HONORS & AWARDS

Dissertation Fellowship Fall 2026

Awarded a departmental Dissertation Fellowship to support accelerated progress on dissertation research in the Fall semester.

WEAI Graduate Student Workshop Summer 2026

Selected for the highly competitive Western Economic Association International's Graduate Student Workshop, focused on job market paper presentation and academic career preparation.

NBER Mentoring Program on Aging and Health Economics Research May 2024

Selected for NBER's mentoring program in aging and health economics supported by the National Institute on Aging.

Graduate Teaching Assistantship 2022 – 2027

Awarded a full tuition scholarship and monthly stipend from the Department of Economics at Colorado State University.

Harley W. Howell Prize May 2022

This award is given to a senior business and economics major for superior academic achievement, promise of leadership career in the field, sound moral character and campus citizenship (issued by the Stead Department of Business Administration & Economics from Coe College).

Spellman Optimum Optimorum May 2022

This award is given to a senior who will be entering an economics Ph.D. program immediately following their graduation from Coe College.

CONFERENCES

Western Economic Association International, *Participant/Presenter* Summer 2026
North American Regional Science Council (NARSC), *Presenter* Fall 2025
NBER Summer Institute Economics of Health Conference, *Participant* Summer 2025
NBER Racial and Ethnic Health Disparities Conference, *Participant* Spring 2024

SERVICES

Brown Bag Seminar Series Aug. 2024 - May 2025
Organizer

PROFESSIONAL MEMBERSHIPS

Columbine Health Systems Center for Healthy Aging at Colorado State University
Western Economic Association International
American Economic Association
American Society of Health Economists

REFERENCES

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